Faculty of Engineering

Module Title: Vibration and Waves

Question Bank

Question (1): Choose the correct answer:

1-The SI base unit of length is:

A) meter B) newton C) joule D) second

2- The SI base unit of heat is:

A) meter B) newton C) joule D) Watt

3- Displacement, velocity, and force are classified as …… quantities.

A) fundamental B) derivative C) scalar D) vector

4- Which is a derived SI unit?

A) kilogram B) second C) newton D) kelvin

5- The dimensions of velocity are:

A) LT^{-1} B) LT^{-2} C) MLT^{-1} D) L^2T^{-1}

6- The dimensions of acceleration are:

A) LT B) LT^{-1} C) LT^{-2} D) MLT^{-2}

7- Force has dimensions:

A) MLT^{-2} B) ML^2T^{-2} C) $ML^{-1}T^{-2}$ D) MT^{-1}

8- Which pair has the same dimensions?

A) work and energy C) pressure and power D) speed and acceleration

9-The dimensions of pressure are:

A) $ML^{-1}T^{-2}$ B) MLT^{-2} C) ML^2T^{-3} D) MT^{-2}

10- A dimensionally valid equation must have:

A) equal numerical values only B) identical dimensions on both sides C) no constants

11- The dimensions of frequency are:

A) T B) T^{-1} C) LT^{-1} D) L^{-1}

12- The dimensions of density are:

A) ML^3 B) ML^{-3} C) $M^{-1}L^3$ D) ML^{-2}

13- The dimensions of energy are:

A) ML^2T^{-2} B) MLT^{-2} C) ML^2T^{-1} D) $ML^{-1}T^{-2}$

14-. The SI unit of momentum is:

A) kg·m/s B) N/m C) J/s D) kg/m^3

15-The dimension of angular frequency is:

A) L B) T^{-1} C) LT^{-1} D) T^2

16- For SHM, the restoring force is:

A) proportional to +x B) proportional to −x C) constant D) zero everywhere

17-The period and frequency satisfy:

A) T=f B) T=1/f C) $T=2\pi f$ D) $T=f^2$

18-……is the motion of an object that regularly returns to a given position after a fixed time interval.

A) Periodic motion B) Simple harmonic motion C) -Unregulated motion

19- At the turning point, the speed is:

A) maximum B) ωA C) zero

20-…… is the maximum displacement in the direction of vibration.

A) Amplitude B) Frequency C) Wavelength

21-An object that completes 100 vibrations in 20 seconds has a frequency of …….

A) 0.2 Hz B)5 Hz C)2000 Hz

22-Increasing mass at fixed k makes the period:

A) shorter B) unchanged C) longer D) zero

23-The SI unit of spring constant is:

A) N·m B) N/m C) kg/m D) J/s

24- When spring extension doubles, its force magnitude:

A) halves B) doubles C) quadruples D) is unchanged

25-……is the distance between two successive points in the wave that have the same phase.

A) - Wavelength B) Wave amplitude C) Frequency

26-Linear mass density is:

A) mass × length B) mass/length C) length/mass D) force/length

27-Doubling string tension changes wave speed by:

A) 2 B) √2 C) 4

28- At a standing-wave node, displacement amplitude is:

A) maximum B) zero C) equal to wavelength

29- For a string fixed at both ends, fundamental wavelength is:

A) L B) 2L C) L/2

30- **……is a disturbance that transfers energy through space or a medium without transporting** matter

A) wave B) Frequency C) Wavelength

31-…….is the phenomenon that occurs when two or more waves meet and overlap in the same medium

A)Periodic Time B) Wave interference C) Wavelength

32-…… is the Interference of two identical sinusoidal waves moving in opposite directions.

A) Wavelength B) Periodic Time C) standing wave.

33-…….is the positions of zero amplitude in standing wave eq.

A)Antinodes B) Nodes C) Wavelength

34-…….is the positions of maximum amplitude in standing wave eq

A) Wavelength B) Antinodes C) Nodes

35- ……is time needed to make one cycle

A) Amplitude B) Frequency C) Periodic Time

(Q2) Classify the following as fundamental or derived quantities:

(a)Length (b) Mass (c)Velocity (d)Force

(Q3) State whether each quantity is scalar or vector:

(a)Time (b) Displacement (c)Speed (d)Acceleration

(Q4)Explain the differences between transverse and longitudinal waves.

(Q5) prove periodic time of oscillating object attached with elastic spring

(Q6) prove Velocity and acceleration of simple harmonic motion

(Q7) prove Energy and Powe transfer by sinusoidal waves on strings

(Q8)Problems:

(1) An object oscillates with simple harmonic motion along the x axis. Its displacement from the origin varies with time according to the equation:

$$X = (4.00\ m)\cos(\pi t + \pi/4)$$

where t is in seconds and the angles in the parentheses are in radians.

(a) Determine the amplitude, frequency, and period of the motion. (b) Determine the maximum speed and maximum acceleration of the object.

(2) A particle oscillates with amplitude 5 cm and angular frequency 4 rad/s. Find its displacement after 0.5 s if it starts from the mean position

(3)A spring has a spring constant of 100N/m and the period of oscillation is 0.628 s .Find the mass attached to the spring

(4)Two masses($m_A = 1\ kg\ \&m_B = 4kg$) are attached separately to identical springs. Find the ratio of their periods

(5) **A block with a mass of 200 g is connected to a light spring for which the force constant is 5 N/m and is free to oscillate on a horizontal, frictionless surface. The block is displaced 5 cm from equilibrium and released from rest (a) Find the period of its motion. (b) Determine the maximum speed of the block. (c) What is the maximum acceleration of the block?**

(6)) A simple pendulum has a time period of 3 s on Earth. Calculate its length taking g= 9.8 m/ s^2

(7) A simple pendulum makes 120 oscillations in 4 minutes. Find

(a)Its periodic time (b) Its frequency (c) Its length ($g = 9.8\frac{m}{s^2}$)

(8) Sinusoidal wave traveling along a string is described by

$$\boldsymbol{Y(x,t) = 0.00327\,sin[\,(72.1(x) - 2.72t)]}$$

Find: Amplitude, wavelength , periodic , wave number , frequency and speed of this wave

(9) A string is under a tension of 200N .Its linear mass density is 0.02 kg/m .Find the speed of the wave

(10) The tension in a string is increased from 100N to 400N .By what factor does the wave speed change?

(11) A string of length 40m has mass 1.2kg and tension 750N .A sinusoidal wave on it has amplitude 3 cm and frequency 25HZ.

Find: linear density , wave speed, angular frequency and average power.

(12) The two waves are

Y_1=4 $\sin(5x - 20t\ )$ Y_2=4 $\sin(5x + 20t\ )$

Find the standing wave equation

(13)

A string fixed at both ends is 8.4 m long and has a mass of 0.12 Kg; it is subjected to a tension of 96 N and starts to oscillate. (a) What is the speed of the waves on the string? (b) What is the longest possible wavelength for the standing waves? (c) Give the frequency of the wave on the string.

(14) A wave has frequency 5 Hz and wavelength 2 m. Find its speed and period.

(15) A traveling wave is y = (0.002 m) sin(20x − 600t), with SI units. Determine amplitude, wavelength, frequency, wave speed and maximum particle speed.

(16) A uniform string has mass 0.3 kg, length 6 m and tension 19.6 N. Find its linear density and traveling-wave speed.

(17) A wave travels at 170 m/s under tension 120 N. What tension is needed for speed 180 m/s on the same string?

(18) A string fixed at both ends has length 1.2 m, mass 0.024 kg and tension 80 N. Find wave speed, fundamental wavelength and fundamental frequency.

(19) A sinusoidal wave travels on a string with $\mu = 0.525$ kg/m, tension 45 N, frequency 120 Hz and amplitude 0.0085 m. Calculate average transmitted power.

(20) A boat produces 12 waves in 20 s. A crest travels 12 m in 6 s and rises 15 cm above equilibrium. Find period, frequency, speed, wavelength and amplitude.

(21) Two waves of equal amplitude 9.8 mm travel in the same direction with phase difference 100°. Find resultant amplitude and classify interference.

(22) A 0.60-kg block attached to a spring with force constant 130 N/m is free to move on a frictionless, horizontal surface. The block is released from rest after the spring is stretched 0.13 m. At that instant, find the force on the block and its acceleration.

(23) Using a small pendulum of length 0.171 m, a geophysicist counts 72 complete swings in a time of 60 s. What is the value of g in this location?

(24) Use dimensional analysis to determine how the linear speed of particle traveling in a circle depends on some or all of the following properties; the radius

of the circle, the angular frequency with which the particle orbits about the circle and the mass of particle.

(25) A spring is mounted horizontally, with its left end fixed. A spring balance attached to the free end and pulled toward the right indicates that the stretching force is proportional to the displacement, and a force of 6 N causes a displacement of 0.03 m. We replace the spring balance with a 0.50-kg glider, pull it 0.02 m to the right along a frictionless air track, and release it from rest. (a) Find the force constant k of the spring. (b) Find the frequency, and period of the resulting oscillation

(26) Show that $x - x_o = v_o t + \frac{1}{2} a t^2$ is dimensionally correct, where x and x_o are the final and initial distances, v_o is the velocity, a is the acceleration and t is the time.

(27) An ultrasonic transducer used for medical diagnosis oscillates at 6.7 MHz. How long does each oscillation take, and what is the angular frequency?

(28) A “second” pendulum is one that moves through its equilibrium position once each second (The period of the pendulum is 2 s). The length of a second’s pendulum is 0.9927 m at Tokyo and 0.9987 m at Cambridge, England. What is the ratio of the free-fall accelerations at these two locations?

(29) Using the method of dimensions, check the validity of: $\boldsymbol{P = P_a + \rho g h}$, where (P) is the pressure at a point placed at a depth (h) from the surface of a fluid of density (ρ), (Pa) is the atmospheric pressure and (g) is gravitational acceleration.

(30) A body is falling from rest under gravity. Find the relation between the distance traveled by the body and between the gravitational acceleration and time elapsed .

(31) Determine the dimensions and SI unit of gravitational constant G from F = GMm/r².

(32) Check the dimensional validity of Q = $Pa^4/(\eta l)$, where Q is volume flow rate, P pressure, a radius, η dynamic viscosity, l length.

(33) A 2 kg block on a horizontal spring (k = 200 N/m) is displaced +0.1 m. Find the restoring force and acceleration.

(34) A vertical spring compresses 2.62 cm under a 4.25 kg mass. Determine its spring constant ($g = 9.8\ m/s^2$).

(35) A pendulum has period 15.5 s on Earth. Estimate its length, then its period on the Moon ($g_{Moon} = 1.67\ m/s^2$).